

Dominik Bittner

dominikbittner2023@gmail.com • Portfolio • Aberdeen, UK

RESEARCH PROFILE

My bond with nature began at an early age, growing up on a farm with its woodlands. I have witnessed firsthand the challenges faced by farmers due to deteriorating soil health, leading to a strong motivation to advance sustainable agricultural practices. Fully in line with UN Sustainable Development Goals, I am particularly invested in helping regions where support is urgently needed, such as in sub-Saharan Africa, where farmers heavily depend on the health of their soils.

EDUCATION

University of Aberdeen <i>PhD in Environmental Sustainability</i>	Sep 2024 – Present Aberdeen, UK
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- Thesis: *Supporting Ethiopian farmers to assess soil and use nature-based solutions with hybrid AI*
- Coursework: Soil science, sustainability, computer vision, interdisciplinary research.

Regensburg University of Applied Sciences <i>M.Sc. Applied Research in Engineering Sciences</i>	Apr 2022 – Feb 2024 Regensburg, GER
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- Thesis: *Investigation into supporting students in learning through eye tracking and deep learning*
- Coursework: Machine learning, deep learning.
- Grade: 1.2 (with distinction)

Regensburg University of Applied Sciences <i>B.Eng. Mechatronics</i>	Oct 2018 – Jul 2022 Regensburg, GER
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- Thesis: *Data augmentation to improve CNN-based semantic segmentation to prevent forest fires*
- Coursework: Software engineering, statistics & optimization.
- Grade: 2.0 (upper second-class)

WORK & RESEARCH EXPERIENCE

University of Aberdeen <i>Research Assistant</i>	Jan 2026 – Present Aberdeen, UK
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- Leading the deployment of an agricultural advisory software for African farmers.

Independent Computational Consultant <i>Freelance Consultant</i>	Nov 2025 – Present Remote
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- Provision of AI expertise for one startup in the agricultural sector.

Regensburg University of Applied Sciences <i>Research Associate</i>	Apr 2022 – Aug 2024 Regensburg, GER
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Fraunhofer Institute for Integrated Circuits IIS <i>Research Intern</i>	Apr 2022 – Dec 2022 Erlangen, GER
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- Analysis of subjects' physiological data to draw conclusions about their affective states.

University of Coimbra
Research Intern

Dec 2021 – Mar 2022
Coimbra, POR

Nottingham Trent University
Research Intern

Aug 2021 – Nov 2022
Nottingham, UK

Maschinenfabrik Reinhausen
Engineering Intern

Sep 2020 – Feb 2021
Regensburg, GER

- Exploration of fibre-optic sensor technology for position detection of vacuum tap-changers.

Siemens AG
Mechatronics Technician

Sep 2013 – Sep 2017
Nuremberg, GER

- Apprenticeship and full-time employment in the production of high voltage converters.

SELECTED PUBLICATIONS

- 2026 **Bittner, D.**, et al. “Assessing the impact of nature-based solutions on soil health in sub-Saharan Africa through farmer-centred methods.” *Environmental Research Letters*, 21(4), 043004.
- 2024 **Bittner, D.**, et al. “Towards use of AI-Powered Hybrid Soil Health Assessment to Design Nature-Based Solutions for Restoration of Degraded Soils in Sub-Saharan Africa.” *AI4SoilHealth Proc.*
- 2024 **Bittner, D.**, et al. “Uncovering Learning Styles through Eye Tracking and Artificial Intelligence”. *ETRA-2024 Proc.*
- 2023 **Bittner, D.**, et al. “Unveiling the Secrets of Learning Styles: Decoding Eye Movements via Machine Learning.” *ICERI Proc.*
- 2023 **Bittner, D.**, et al. “In-depth Benchmarking of Transfer Learning Techniques for Improved Bottle Recognition.” *ICPRS Proc.*
- 2023 **Bittner, D.**, et al. (2023). “Towards Eye Tracking based Learning Style Identification.” *ECSEE Proc.*
- 2022 **Bittner, D.**, et al. “Generating synthetic multispectral images for semantic segmentation in forestry applications.” *ICRA Proc.*

CONFERENCES & INVITED TALKS

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| Dec 2025 | Poster | “Hybrid Soil Engine“ | <i>BSSS Annual Conf., Manchester, UK</i> |
| Dec 2025 | Oral | “Hybrid Soil Engine“ | <i>BSSS EC Conf., Manchester, UK</i> |
| Oct 2025 | Oral | “Hybrid Soil Engine“ | <i>NESEN Conf., Aberdeen, UK</i> |
| Oct 2025 | Poster | “Hybrid Soil Engine“ | <i>Interdisc. Symp., Aberdeen, UK</i> |
| Dec 2024 | Poster | “Hybrid Soil Engine“ | <i>AI4SoilHealth Conf., Budapest, HUN</i> |
| Jun 2024 | Oral | “DL in Eye Tracking“ | <i>ETRA Conf., Glasgow, UK</i> |
| Dec 2023 | Invited | “DL in Education“ | <i>Univ. of Vienna, AUT</i> |
| Nov 2023 | Oral | “DL in Eye Tracking“ | <i>ICERI Conf., Seville, SPA</i> |
| Jul 2023 | Oral | “DL Benchmarking“ | <i>ICPRS Conf., Guayaquil, ECU</i> |
| Jun 2023 | Oral | “DL in Education“ | <i>ECSEE Conf., Seon, GER</i> |
| May 2022 | Oral | “DL in Forestry“ | <i>ICRA Conf., Philadelphia, USA</i> |

GRANTS & AWARDS

ISMC travel award <i>Travel grant to attend International Soil Modeling Consortium conference</i>	Sep 2026 520 GBP
SUSTAIN CDT Doctoral Stipend <i>Full research funding and maintenance</i>	Sep 2024 – Aug 2028 94,000 GBP
Bavarian State Ministry Grant <i>PI of the eye movement modelling examples project (No. L.1-H2493.0/15)</i>	Apr 2024 – Apr 2025 52,000 GBP
DAAD HAW International Stipend <i>International scholarship for thesis completion abroad</i>	Dec 2021 – Mar 2022 4,300 GBP
DAAD RISE Programme Stipend <i>International research scholarships for science internships</i>	Aug 2021 – Nov 2021 3,500 GBP

TEACHING & SUPERVISION EXPERIENCE

University of Aberdeen <i>Demonstrator & Supervisor</i>	Sep 2024 – Present <i>Aberdeen, UK</i>
■ Courses: AI, Evaluation of AI Systems, Statistical Analysis of Biological Data ■ Supervisees: Jeanne Biegel, Adam Ponting, Daniel Eze, Olubusuyi Omowaye	

SKILLS

Research: Experimental design, qualitative & quantitative research.

Computer Science: Scientific ML & DL, Computer Vision, Python, R, Julia.

Soil Science: Soil classification, soil analysis, sustainability.

Transferable: Academic writing, public speaking, teamwork, leadership.

Languages: German (Native), English (Professional Working Proficiency), Spanish (Basics).

EXTRACURRICULAR, SERVICE & MEMBERSHIPS

- **Board member:** Network coordinator at Aberdeen-Africa Research Network.
- **Student representative:** The students' voice in the SUSTAIN-CDT programme.
- **Smoke diver:** Trained in high-stress rescue operations at local voluntary fire brigade.
- **Coach and board member:** Support and guidance for the local youth football club.
- **Memberships:** British Society of Soil Science, International Soil Modeling Consortium

INTERESTS

- **Run the farm:** The family farm is a lifelong, shared commitment and a round-the-clock passion.
- **In the orchestra:** Playing solo is great, in a band, it's all about listening and connecting.

REFEREES

Professor Jo Smith
University of Aberdeen
jo.smith@abdn.ac.uk
Dr. David Portugal
University of Coimbra
davidbsp@isr.uc.pt

Professor Jürgen Mottok
Regensburg University of Applied Sciences
juergen.mottok@oth-regensburg.de